DOCKER

Docker is an open-source containerization platform used to develop, package, deploy, and run applications inside lightweight containers. Containers include the application code, dependencies, libraries, and runtime environment required for execution.

DOCKER IMAGE

- A template or blueprint used to create Docker containers.
- Contains application code, libraries, dependencies, and configurations.

DOCKER CONTAINER

- A lightweight and isolated environment used to run applications.
- Runs applications consistently with all required dependencies.

DOCKER USES

- **Application Deployment -** Deploys applications quickly using containers.
- **Containerization -** Packages application code and dependencies together.
- **DevOps Automation** - Supports CI/CD and automated deployment workflows.
- **Environment Consistency -** Runs applications the same in development, testing, and production.
- **Microservices Support -** Runs multiple services independently in separate containers.

DOCKER FEATURES

- **Lightweight Containers -** Uses fewer system resources than virtual machines.
- **Portability** - Containers run on any system with Docker installed.
- **Isolation -** Each container works independently without affecting others.
- **Fast Startup -** Containers start and stop quickly.
- **Image Management -** Docker images can be created, stored, and shared easily.

DOCKER COMMANDS

- **docker images**

Displays all Docker images available in the system.

- **docker pull httpd**

Downloads the Apache HTTPD Docker image.

- **docker run -itd --name cont1 -p 8080:80 httpd:latest**

Creates and runs an HTTPD container.

- **docker ps**

Displays running Docker containers.

- **docker ps -a**

Displays all Docker containers including stopped containers.

- **docker stop <container-id>**

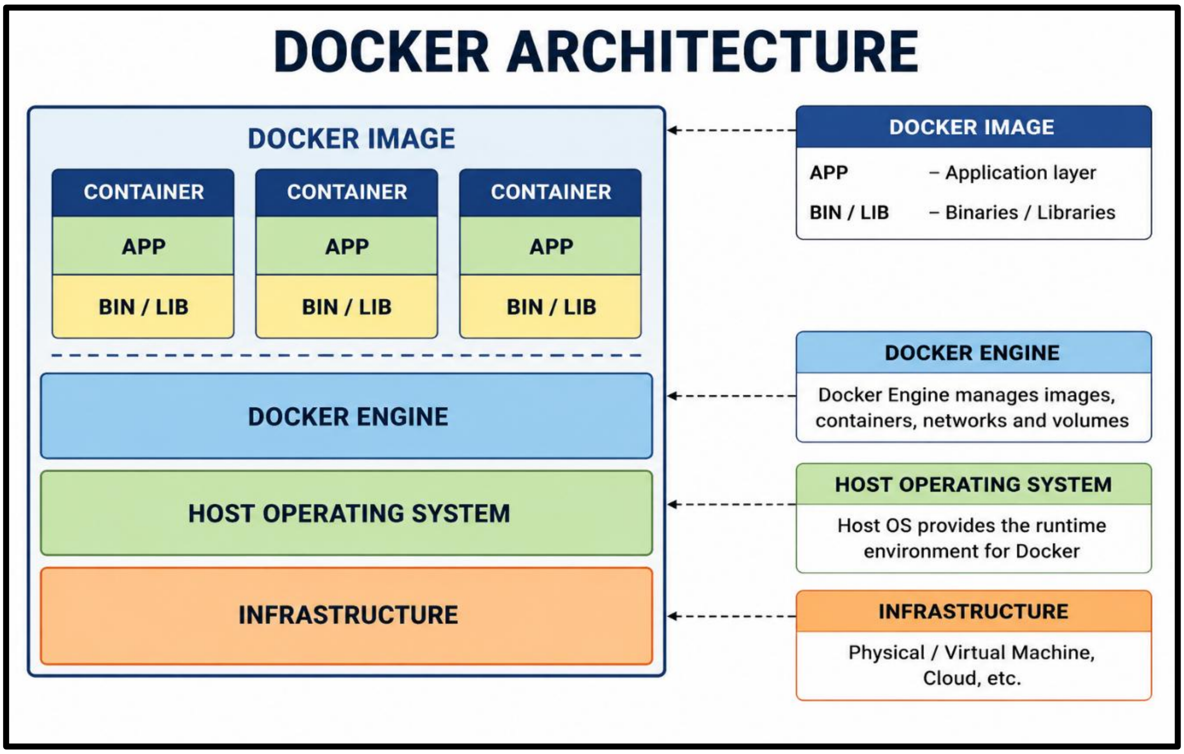
Stops a running Docker container.

- **docker rm <container-id>**

Removes Docker containers.

- **docker rmi nginx:latest httpd:latest**

Removes Docker images from the system.



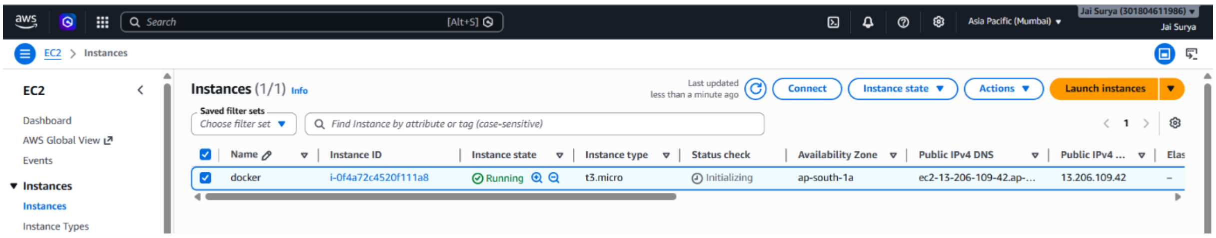
HANDS-ON FOR DOCKER CONTAINERIZATION

Screenshot: Create and Connect Linux Instance

• Launch a Linux EC2 instance in AWS

• Select the instance and click Connect

• Connect to the Linux instance successfully



Screenshot: Update Linux Instance Packages

• Execute sudo su

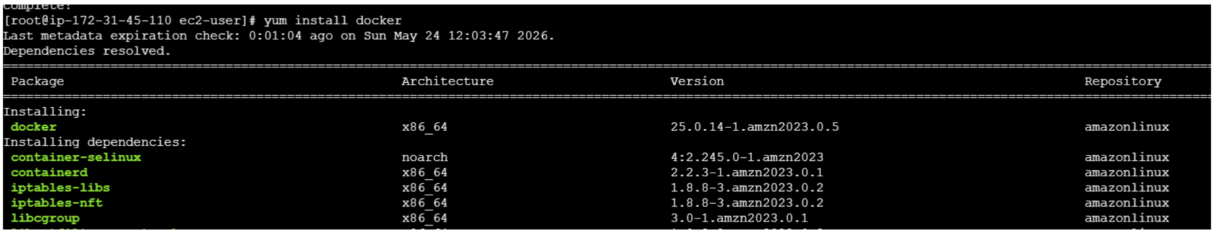
• Execute yum update -y

• System packages updated successfully

Screenshot: Install Docker in Linux Instance

• Execute yum install docker -y

• Docker installed successfully in the Linux EC2 instance



Screenshot: Start and Verify Docker Service

• Execute systemctl start docker

• Execute systemctl enable docker

- Execute systemctl status docker
- Docker service started and enabled successfully

```
[root@ip-172-31-45-110 ec2-user]# systemctl start docker
[root@ip-172-31-45-110 ec2-user]# systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[root@ip-172-31-45-110 ec2-user]# systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-05-24 12:06:33 UTC; 20s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 27583 (dockerd)
      Tasks: 9
```

Screenshot: Pull Docker Image

- Execute docker images
- No Docker images available initially
- Execute docker pull httpd
- Apache HTTPD Docker image downloaded successfully

```
[root@ip-172-31-45-110 ec2-user]# docker images
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
[root@ip-172-31-45-110 ec2-user]# docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
5b4d6ff92fc4: Pull complete
f086d7c0b862: Pull complete
4f4fb700ef54: Pull complete
f60b6f23733a: Pull complete
c3f994ae0cac: Pull complete
ed804d184361: Pull complete
Digest: sha256:2a7deaaaf357261a1dffcb8fc725f4aaba2af95fbde1a40a68bca7cb0f03594e
Status: Downloaded newer image for httpd:latest
docker.io/library/httpd:latest
[root@ip-172-31-45-110 ec2-user]#
```

Screenshot: Verify Downloaded Docker Images

- Execute docker images
- Downloaded httpd Docker image displayed successfully

```
[root@ip-172-31-45-110 ec2-user]# docker images
REPOSITORY   TAG       IMAGE ID       CREATED      SIZE
httpd        latest    b9097c67d4a2   4 days ago   117MB
[root@ip-172-31-45-110 ec2-user]#
```

Screenshot: Run Docker Container

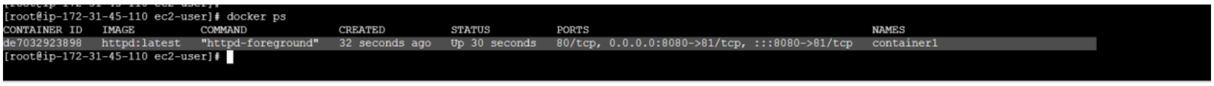
- Execute docker run -itd --name cont1 -p 8080:80 httpd:latest
- Docker container cont1 created and running successfully using the HTTPD image

```
[root@ip-172-31-45-110 ec2-user]# docker run -itd --name container1 -p 8080:81 httpd:latest
de7032923898e8eee4a3340a0fbce5ab5e86500b5059326564bc698f0da64d1b
[root@ip-172-31-45-110 ec2-user]#
```

Screenshot: Verify Running Docker Containers

• Execute docker ps

• Running Docker container cont1 displayed successfully with container ID, status, and port mapping



Screenshot: Access Docker Container in Browser

• Copy the public IP address of the EC2 instance

• Open browser and enter Public-IP:8080

• Apache HTTPD default web page loaded successfully from the Docker container

Screenshot: Pull and Run Nginx Docker Container

• Execute docker pull nginx

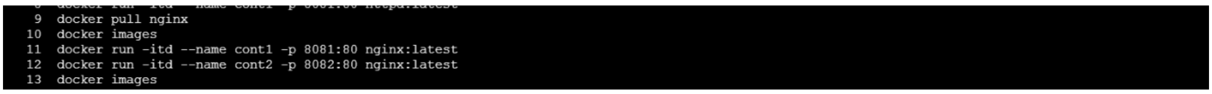
• Nginx Docker image downloaded successfully

• Execute docker images

• HTTPD and Nginx images displayed successfully

• Execute docker run -itd --name cont2 -p 8081:80 nginx:latest

• Nginx Docker container created and running successfully

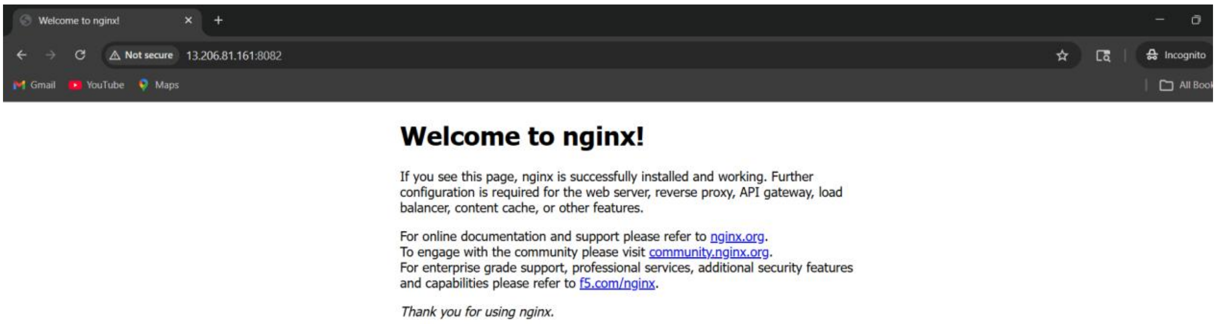


Screenshot: Access Nginx Container in Browser

• Copy the public IP address of the EC2 instance

• Open browser and enter Public-IP:8081

• Nginx default web page loaded successfully from the Docker container



Screenshot: Verify All Docker Containers

• Execute docker ps

• Displays currently running Docker containers

• Execute docker ps -a

• Displays all Docker containers including running and stopped containers

```
[root@ip-172-31-40-238 ec2-user]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS                                   NAMES
6e8bcd376c45   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8082->80/tcp, :::8082->80/tcp   cont2
bb04eeab6215   httpd:latest   "httpd-foreground"       3 minutes ago        Up 3 minutes        0.0.0.0:8081->80/tcp, :::8081->80/tcp   cont1
[root@ip-172-31-40-238 ec2-user]# docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS                                   NAMES
6e8bcd376c45   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8082->80/tcp, :::8082->80/tcp   cont2
bb04eeab6215   httpd:latest   "httpd-foreground"       3 minutes ago        Up 3 minutes        0.0.0.0:8081->80/tcp, :::8081->80/tcp   cont1
[root@ip-172-31-40-238 ec2-user]#
```

Screenshot: Stop Docker Containers

• Execute docker stop <cont1-container-id>

• Execute docker stop <cont2-container-id>

• Both Docker containers stopped successfully

• Execute docker ps

• No running Docker containers displayed successfully

```
[root@ip-172-31-40-238 ec2-user]# docker stop 6e8bcd376c45
6e8bcd376c45
[root@ip-172-31-40-238 ec2-user]# docker stop bb04eeab6215
bb04eeab6215
[root@ip-172-31-40-238 ec2-user]# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[root@ip-172-31-40-238 ec2-user]#
```

Screenshot: Verify Stopped Docker Containers

• Execute docker ps -a

• Displays all Docker containers including stopped containers cont1 and cont2 successfully

```
[root@ip-172-31-40-238 ec2-user]# docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS                          PORTS     NAMES
6e8bcd376c45   nginx:latest   "/docker-entrypoint.…"   5 minutes ago   Exited (0) About a minute ago             cont2
bb04eeab6215   httpd:latest   "httpd-foreground"       7 minutes ago   Exited (0) About a minute ago             cont1
```

Screenshot: Verify Container Removal

- Execute docker rm <cont1-container-id> <cont2-container-id>
- Docker containers removed successfully
- Execute docker ps -a
- No Docker containers displayed successfully

```
[root@ip-172-31-40-238 ec2-user]#
[root@ip-172-31-40-238 ec2-user]# docker rm 6e8bcd376c45 bb04eeab6215
6e8bcd376c45
bb04eeab6215
[root@ip-172-31-40-238 ec2-user]# docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[root@ip-172-31-40-238 ec2-user]#
```

Screenshot: Remove Docker Images

- Execute docker images
- Displays available Docker images
- Execute docker rmi nginx:latest httpd:latest
- Nginx and HTTPD Docker images removed successfully

```
[root@ip-172-31-40-238 ec2-user]# docker images
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
nginx        latest    7aaca76c508f   42 hours ago   161MB
httpd        latest    b9097c67d4a2   4 days ago     117MB
[root@ip-172-31-40-238 ec2-user]#
[root@ip-172-31-40-238 ec2-user]# docker rmi nginx:latest httpd:latest
Untagged: nginx:latest
Untagged: nginx@sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Deleted: sha256:7aaca76c508f7d121ff29cbe9dd071012486d00c21e17655eb1a1dfb711e9330
Deleted: sha256:7f86f065931373875d541bf5863fa17ac1adeb4ebf3027ac3a8ed3cde9d4c710
Deleted: sha256:7a9c00f83cfaf7ff9ca442bf1c6c91b24a36f79d1154ef43272f05abd7b1ecbe
Deleted: sha256:842055d2621005134843fa4cb8d7a5540af9a525e25c46a8c1681a5516b40a23
Deleted: sha256:e572894f043bbca5b7f8ff20c08624ac66dc120184941fab812a3254c6d74525
Deleted: sha256:26abf6b97663dca0c83230e36673baf6a91b82af1d590125037abe33a2539f5a
Deleted: sha256:db8505aeeb10636b6cddc9a39b975feb4ac3afbdb87927a51c5117c2f5d09448
Untagged: httpd:latest
Untagged: httpd@sha256:2a7deaaaf357261a1dffcb8fc725f4aaba2af95fbde1a40a68bca7cb0f03594e
Deleted: sha256:b9097c67d4a2c6d267e015e6297a61afdd1870f904f9229fce3c068db860cd68
Deleted: sha256:6e463bbf010f913986952d453ddb9b9ccc6156c38b45299d4cf54ad07294234a
Deleted: sha256:e2940635c5b95917ffa196896a67a50a0c7ac739722db71591bef27be119802e
Deleted: sha256:a845eb309d6f115a9e9d7d2a100f82b6867ef8b0572412ed8db6188f4ee537b7
Deleted: sha256:2e5669539e9ee3c7437ef60680d1c01e24013dceb5f48d2dee84c5584aaf32be
Deleted: sha256:0b48ec94efabe9e8ecfaef9edfa37d1820260172b7765b6c6494ab252c755fb9
Deleted: sha256:219a998c60509502b47b97f1158067d5dd62640d2d689560d32cfd5594f6bc40
[root@ip-172-31-40-238 ec2-user]#
```

Screenshot: Verify Docker Images Removal

- Execute docker images
- No Docker images displayed successfully after image removal

```
[root@ip-172-31-40-238 ec2-user]# docker images
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
[root@ip-172-31-40-238 ec2-user]#
```